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Final Project

15.465 Alphanomics

Assigned data file: `fmreg_data_std_28.csv`

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Executive summary

We recommend longing Estée Lauder (EL) and shorting Coty (COTY). The comparison set is three U.S.-listed beauty names—EL, e.l.f. Beauty (ELF), and Coty (COTY)—using nine significant signals from our Part 1 multivariate stack, April 2026 characteristics, weights proportional to each signal’s $|t|$, and within-trio ranks on each line item. EL ranks at the top because it leads on market-adjusted momentum and SUE, posts the lowest realized daily volatility (favorable under our negative volatility loading), and avoids the weakest insider-selling rank among the three under the Part 1 Form 4 screen (open-market officer trades; gifts excluded). Coty lands at the bottom because it ranks last on momentum, earnings surprise, and SUE and scores worst on the F-score tier, which more than offsets its relative strength on book-to-market and insider flow where the model assigns less weight. e.l.f. Beauty sits between EL and Coty on the composite; we treat it as neutral rather than as a directional leg of the pair. Methodology and full supporting tables follow below.

Part 1: Data Analysis

Step A: Univariate Analysis

For each month, we run a cross-sectional univariate regression of monthly return (`ret`) on one standardized signal:

$$\text{ret}_{i,t} = \alpha_t + \beta_t \cdot \text{signal}_{i,t} + \varepsilon_{i,t}.$$

We repeat this separately for each of the 10 signals and for all available months. Following the assignment instructions, we compute Fama-MacBeth standard errors as:

$$SE(\bar{\beta}) = \frac{\text{std}(\beta_t)}{\sqrt{T}}, \quad t = \frac{\bar{\beta}}{SE(\bar{\beta})}.$$

We run these monthly cross-sectional regressions across all available months and save the full time series of coefficients for each signal for downstream analysis.

The figure above reports the coefficient paths over time. The table below reports the time-series average coefficients, Fama-MacBeth t-statistics, average R-squared values, and the number of observations used.

Table 1. Univariate Fama-MacBeth summary statistics by signal.

Signal	Avg coef	SE	t stat	Avg R ²	Months	Obs	Sig. at 10%
inssell	-0.0247	0.0039	-6.3114	0.0911	60	12,000	Yes
booktomarket	0.0041	0.0015	2.8095	0.0064	60	12,000	Yes
momentum	0.0170	0.0021	8.0392	0.0433	60	12,000	Yes
grossprofit	0.0062	0.0020	3.1563	0.0135	60	12,000	Yes
earnsurprise	0.0122	0.0015	8.0263	0.0137	60	12,000	Yes
sue	0.0123	0.0018	6.8555	0.0185	60	12,000	Yes
fscore	0.0047	0.0016	2.9112	0.0107	60	12,000	Yes
xfin	-0.0063	0.0017	-3.8180	0.0099	60	12,000	Yes
volatility	-0.0135	0.0020	-6.6568	0.0195	60	12,000	Yes
accruals	0.0023	0.0020	1.1736	0.0104	60	12,000	No

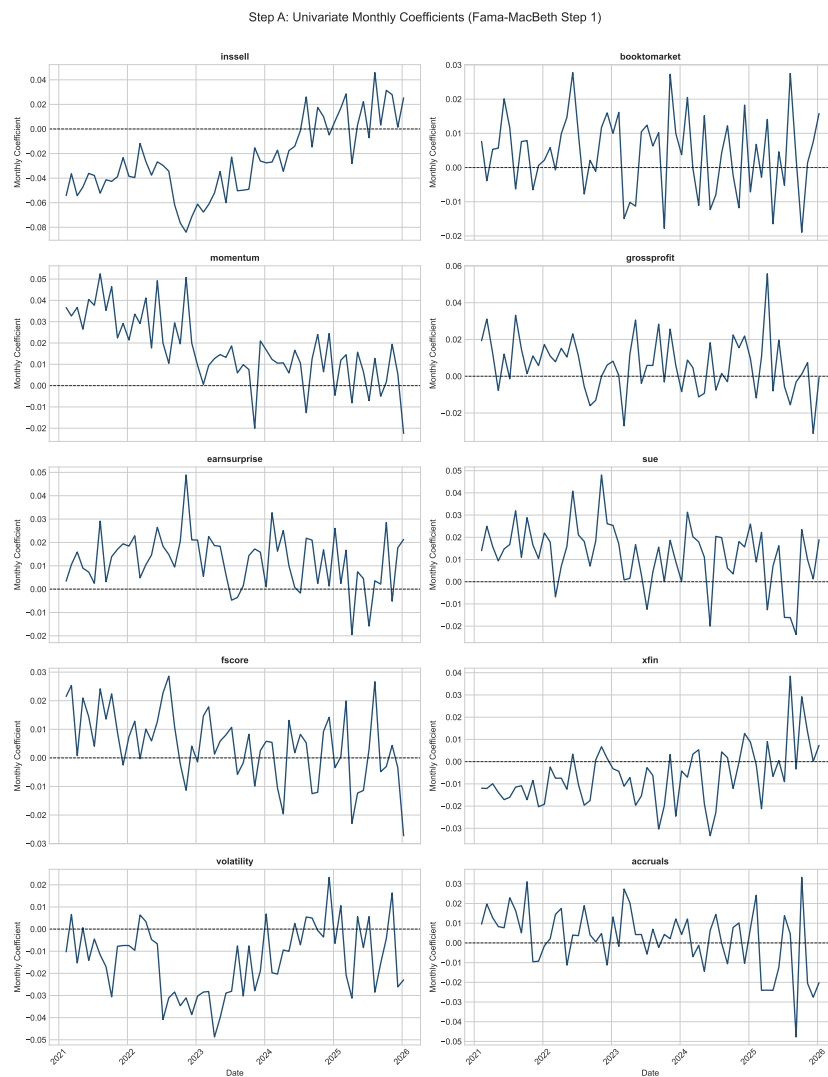


Figure 1. Monthly univariate slope coefficients by signal (5x2 grid).

The next table reports the share of months in which each signal's monthly coefficient is positive.

Table 2. Fraction of months with positive univariate coefficient.

Signal	Months positive	Months used	Fraction positive
inssell	14	60	0.2333
booktomarket	39	60	0.6500
momentum	53	60	0.8833
grossprofit	40	60	0.6667
earnsurprise	54	60	0.9000
sue	53	60	0.8833
fscore	40	60	0.6667
xfin	18	60	0.3000
volatility	13	60	0.2167
accruals	38	60	0.6333

In the univariate tests, the strongest relationships with monthly returns are momentum, earnings surprise, SUE, and volatility based on absolute t-statistics and stable coefficient signs through time. These signals combine both statistical strength and directional consistency: momentum (avg coef=0.0170, t-stat=8.04, fraction positive=88.33%); earnsurprise (avg coef=0.0122, t-stat=8.03, fraction positive=90.00%); sue (avg coef=0.0123, t-stat=6.86, fraction positive=88.33%); volatility (avg coef=-0.0135, t-stat=-6.66, fraction positive=21.67%).

Insider selling also stands out economically because it has the largest absolute average slope (-0.0247) and is negative in most months (only 23.33% of months are positive). A practical interpretation is that higher net insider selling is associated with lower subsequent returns, which supports using insider-selling intensity as a short-side ranking input rather than a standalone timing signal.

At the weak end, accruals has a low t-statistic (1.17) and limited reliability, so we would treat it as low-conviction in this sample. Across the coefficient paths, signs are generally more stable than magnitudes, which is typical in noisy monthly cross-sections where signal strength varies with market regime.

Economically, a coefficient gives the expected change in monthly return for a one-standard-deviation change in a signal (because signals are standardized within month). For example, a momentum slope near 0.0170 means a stock that is 1 SD higher in momentum is expected to have roughly 0.0170 higher monthly return, holding only the univariate setup fixed.

A plausible mechanism for momentum is slow information diffusion and trend-chasing behavior, where investors continue to ride recent winners before fully updating valuations. For insider selling, one explanation is that by the time trades are disclosed, part of the informational edge is already reflected in prices; nevertheless, persistent net selling still contains incremental negative information in this dataset.

Step B: Multivariate Analysis

We next run monthly cross-sectional regressions including all ten signals simultaneously, so each coefficient reflects a partial relationship with return while controlling for the other signals.

The figure reports the multivariate coefficient paths over time. The table below reports the time-series average coefficients, Fama-MacBeth t-statistics, average R-squared values, and the number of observations used.



Figure 2. Monthly multivariate slope coefficients by signal (5x2 grid).

Table 3. Multivariate Fama-MacBeth summary statistics by signal.

Signal	Avg coef	SE	t stat	Avg R ²	Months	Obs	Sig. at 10%
inssell	-0.0247	0.0039	-6.3052	0.2139	60	12,000	Yes
booktomarket	0.0046	0.0016	2.8890	0.2139	60	12,000	Yes
momentum	0.0171	0.0022	7.7881	0.2139	60	12,000	Yes
grossprofit	0.0060	0.0019	3.2374	0.2139	60	12,000	Yes
earnsurprise	0.0098	0.0018	5.5437	0.2139	60	12,000	Yes
sue	0.0055	0.0023	2.4416	0.2139	60	12,000	Yes
fscore	0.0042	0.0015	2.8494	0.2139	60	12,000	Yes
xfin	-0.0075	0.0016	-4.7232	0.2139	60	12,000	Yes
volatility	-0.0134	0.0020	-6.6122	0.2139	60	12,000	Yes
accruals	0.0022	0.0019	1.1838	0.2139	60	12,000	No

In the multivariate tests, most signals remain significant at the 10% threshold after controlling for the full signal set. The strongest multivariate predictors by absolute t-statistic are momentum (avg coef=0.0171, t-stat=7.79); volatility (avg coef=-0.0134, t-stat=-6.61); insell (avg coef=-0.0247, t-stat=-6.31); earnsurprise (avg coef=0.0098, t-stat=5.54). Accruals remains weak and statistically insignificant (t-stat=1.18).

The largest drop in t-statistic versus univariate appears in sue (change -4.41), which suggests overlap with other signals rather than pure standalone information loss. More broadly, when coefficients shrink after controls, the interpretation is that part of the univariate relation was shared with correlated characteristics.

The average multivariate R-squared is about 0.214, which is meaningful for monthly cross-sectional stock-return variation: the model explains a nontrivial share of cross-firm return differences, but still leaves substantial residual noise, so risk controls and diversification remain necessary.

In economic terms, each multivariate coefficient is a partial effect: a one-standard-deviation increase in a signal corresponds to the coefficient change in monthly return, holding all other signals fixed. This is the right lens for portfolio construction because it reflects incremental contribution rather than raw standalone association.

Step C: Sharpe Ratio Analysis

Using the Step B t-statistics, we compute implied Sharpe ratios and report them in descending order.

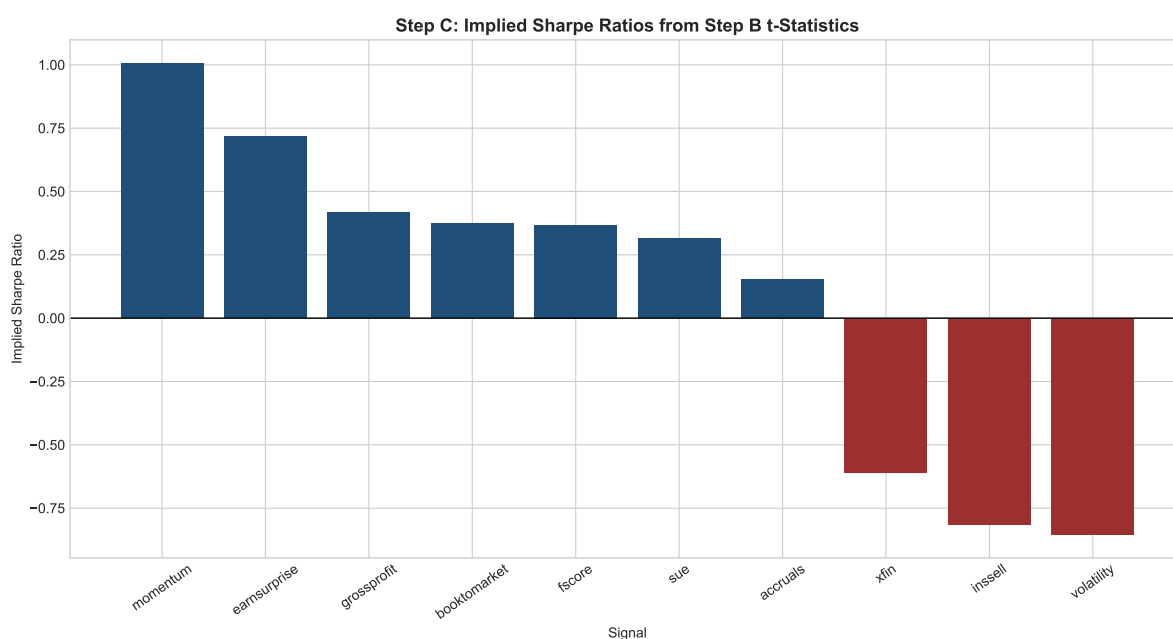


Figure 3. Implied Sharpe ratios by signal, sorted in descending order.

Using the assignment convention, implied Sharpe ratios are computed as t-stat divided by the square root of the number of months. The strongest four signals in absolute implied Sharpe terms are momentum (implied Sharpe=1.0054); volatility (implied Sharpe=-0.8536); insell (implied Sharpe=-0.8140); earnsurprise (implied Sharpe=0.7157). This ranking highlights both high positive and high negative opportunities.

Table 4. Implied Sharpe ratios from Step B t-statistics.

Signal	t stat	Months	Implied Sharpe
momentum	7.7881	60	1.0054
earnssurprise	5.5437	60	0.7157
grossprofit	3.2374	60	0.4179
booktomarket	2.8890	60	0.3730
fscore	2.8494	60	0.3679
sue	2.4416	60	0.3152
accruals	1.1838	60	0.1528
xfin	-4.7232	60	-0.6098
inssell	-6.3052	60	-0.8140
volatility	-6.6122	60	-0.8536

For strategy design, strongly positive implied Sharpe signals are natural long candidates, while strongly negative implied Sharpe signals can be used as short candidates or negative tilts. In practice, we would combine sign, magnitude, and reliability into position sizing rather than using equal weights.

A robust implementation would include turnover and trading-cost controls, sector/market-neutral constraints, and conservative risk budgeting to avoid overexposure to a single theme despite strong historical signal performance.

Part 2: Research and Recommendation

Step A: Signal Collection and Measurement

Of the ten signals examined in Part 1, nine were statistically significant at the 10% level in the multivariate Fama MacBeth tests ($|t| > 1.645$); only accruals ($t \text{ stat} = 1.18$) was insignificant and is therefore excluded from Part 2. We collected the most recent available value of each remaining signal as of April 24, 2026 for our three pre registered companies: The Estée Lauder Companies Inc. (EL), Coty Inc. (COTY), and e.l.f. Beauty, Inc. (ELF).

For each signal we replicated the Part 1 definition as closely as possible using publicly available data. Sources are listed below.

- **Insider Selling (inssell):** We accessed EDGAR and downloaded every Form 4 insider trading filing for each company over the trailing three months (window: January 23, 2026 to April 24, 2026). We parsed each filing and kept only the four officer roles specified in Part 1 (CEO or President, CFO or Finance Director, COO, and Board Directors) and only genuine open market buys (transaction code P) and sells (transaction code S). We excluded RSU vestings, equity awards, option exercises, tax withholding (code F), and gifts (code G). After this filter we were left with 0, 3, and 2 qualifying transactions for EL, ELF, and COTY respectively. We computed inssell as (shares sold minus shares bought) divided by the firm's most recent shares outstanding.
- **Book to Market:** Book equity (total assets minus total liabilities) from each company's most recent SEC quarterly or annual report, divided by current market capitalisation (Yahoo Finance closing price on April 24, 2026 multiplied by shares outstanding from the same filing).
- **Momentum:** Cumulative market adjusted return over the trailing 12 months, excluding the most recent month to avoid short term reversal, computed from Yahoo Finance daily total return prices using the S&P 500 (ticker GSPC) as the market benchmark.
- **Gross Profitability:** Gross profit (revenue minus cost of goods sold) divided by total book assets, both taken from the most recent annual report (Form 10K) via WRDS Compustat.
- **Earnings Surprise:** Take the earnings per share the company just reported, subtract the average analyst forecast for that same quarter (from I/B/E/S on WRDS), and divide by the stock price five trading days before the announcement (price from Yahoo Finance). The reported EPS side comes from Compustat on WRDS, which is built from the company's SEC quarterly filing.
- **SUE:** The SUE signal asks whether the company earned more this quarter than it did in the same quarter a year ago, scaled to be comparable across companies. We pulled diluted quarterly EPS from each company's most recent quarterly report and from its quarterly report one year earlier (both via WRDS Compustat). SUE is then computed as (latest quarter actual EPS minus same quarter prior year actual EPS) divided by the closing stock price five trading days before the latest announcement (price from Yahoo Finance). The earnings announcement date itself also comes from WRDS Compustat.
- **Piotroski F Score:** A nine criterion binary composite covering profitability, leverage and liquidity, and operating efficiency, computed from the most recent and prior year annual

reports via WRDS Compustat.

- External Financing (xfin): Net financing activities, defined as (debt issued plus equity issued) minus (debt repaid plus shares repurchased), divided by total assets. All inputs come from the most recent annual report cash flow statement via WRDS Compustat.
- Volatility: Standard deviation of daily stock returns over the trailing 12 months, computed from Yahoo Finance daily prices.

Table 5 reports the raw value of each signal for each company. All values are reported in decimal form to be consistent with Part 1. The F Score is an integer between 0 and 9.

Table 5. Signal values by company, as of April 24, 2026.

Signal	EL	ELF	COTY	Source
Insider Selling (inssell)	0.0000	0.0008	-0.0001	EDGAR Form 4 (Jan 23 to Apr 24, 2026)
Book to Market	0.1432	0.2968	1.8523	Latest 10Q and Yahoo Finance close
Momentum (12m ex. last)	0.0298	-0.2066	-0.7882	Yahoo Finance daily prices
Gross Profitability	0.5744	0.7850	0.3406	Latest 10K via WRDS Compustat
Earnings Surprise	-0.0033	-0.0005	-0.1029	IBES (WRDS) and Yahoo Finance
SUE	0.0182	0.0041	-0.0514	WRDS Compustat and Yahoo Finance
Piotroski F Score	5	5	4	Latest and prior 10K, WRDS Compustat
External Financing (xfin)	-0.0575	-0.0579	-0.0113	Latest 10K cash flow statement, WRDS
Volatility (daily)	0.0294	0.0443	0.0317	Yahoo Finance daily returns

Deviation from Part 1. Part 1 signals were standardized within each calendar month across 200 firms. With only three firms here, a within-month z -score is not informative, so values above are reported in raw decimal form (the F Score is an integer 0–9). Step B then converts these raw values into within-trio rank scores, which is the appropriate analogue to a z -score in a small cross-section.

As-of dates by company. EL: latest 10K covers fiscal year ending June 30, 2025; latest 10Q covers the quarter ending December 31, 2025; latest earnings announcement was on February 5, 2026. COTY: latest 10K covers fiscal year ending June 30, 2025; latest 10Q covers the quarter ending December 31, 2025; latest earnings announcement was on February 5, 2026. ELF: latest 10K covers fiscal year ending March 31, 2025; latest 10Q covers the quarter ending December 31, 2025; latest earnings announcement was on February 4, 2026. All measurement dates and underlying data files are recorded in the project repository under DATA/raw_wrds and DATA/inssell.

Table 5 reports the nine signal values for our three companies as of April 24, 2026. We encountered a few judgement calls and edge cases worth flagging, grouped into three themes: missing data, ambiguous classifications in the insider trading filings, and fiscal calendar mismatches.

Genuine missing values were rare. The main convention worth flagging is Part 1’s explicit “Zero if missing” rule for Insider Selling, which we followed. Estée Lauder is the clearest example: over the trailing three months we found 21 Form 4 filings on EDGAR, but after filtering for the four officer roles named in Part 1 (CEO, CFO, COO, and Board Directors) and for actual market trades, zero qualifying transactions remained. Estée Lauder’s Insider Selling is therefore reported as zero, which is the correct treatment under the Part 1 rule rather than a true data gap. For External Financing, Compustat occasionally leaves the share repurchase or debt issuance line items blank in years where the company did not undertake those activities. We treated those blanks as zero, since a missing line item in a 10K cash flow statement almost always means the activity did not happen rather than that the data was lost. For Earnings Surprise the idea is simple. Before the company reports quarterly earnings, outside analysts publish their guesses for earnings per share. After the report, we know the true number.

Earnings Surprise compares those two: it measures how much the reported earnings per share differed from the average analyst guess, scaled by the stock price shortly before the news. We read the reported number from the company's quarterly filing as it appears in Compustat on WRDS. We read the average analyst guess from I/B/E/S on WRDS, which is the usual place academics store those forecasts. We only keep guesses that were already in the database before the announcement date, so we are not using information that only appeared after the company spoke. That gives one number per firm in the signal table. For EL, COTY, and ELF the I/B/E/S feed had a guess we could use for the quarter in the table, so we did not treat this signal as missing.

The Insider Selling signal required the most interpretation work because the underlying Form 4 data is rich but messy. Three calls are worth describing.

First, we excluded family stock gifts. In April 2026, Estée Lauder filed two Form 4 reports describing a gift of 4,768,846 Class B shares from Ronald S. Lauder to Aerin Lauder. This is family estate planning rather than a market view about the firm's prospects, so we did not count it as either a buy or a sell. The transaction code G used for gifts is the standard Form 4 marker for this kind of activity, and dropping it is consistent with the spirit of the Part 1 definition, which targets buys and sells as information about insider conviction.

Second, we excluded the option exercise leg of compound transactions. ELF's CEO Tarang Amin exercised stock options for 71,000 shares on April 1, 2026 (transaction code M) and on the same day sold 52,880 of the resulting shares in the open market (transaction code S). The 71,000 share leg is the option exercise itself, not a market trade, so we counted only the 52,880 shares actually sold. Counting the M leg would have understated the CEO's true cash out from the position.

Third, Coty was a little messy for Insider Selling. In our window no one who looks like a classic group CEO was buying or selling stock in the open market. The interim CEO only showed up with automatic stock awards, not open market trades. The only open market purchases we saw were from the CFO and from a president who runs one business division, not the whole firm. Part 1 still treats some presidents like a CEO, so we kept that division president in the count. The ranking barely moves either way: Coty's insider selling number rounds to about zero (-0.0001 with him in, basically zero if we drop him), so Step B does not hang on this call.

Our three companies do not share the same fiscal calendar. Estée Lauder and Coty close their fiscal year on June 30, while ELF closes on March 31. As a result, the most recent annual 10K is dated June 30, 2025 for the first two and March 31, 2025 for ELF. This affects every signal that draws on annual data, namely Gross Profitability, External Financing, and the F Score, and means each firm's annual signals are anchored to slightly different dates, with ELF's being roughly three months staler than the other two. Part 1 points to the most recent fiscal year, so we took gross profit, balance sheet, and cash flow inputs straight from each firm's latest annual report (Form 10K). We did not rebuild a one year total by adding up the last four quarterly reports, and we did not take a single quarter from a Form 10Q and multiply by four to pretend it was a full year. Those shortcuts show up sometimes in industry, but they are not the same object as the full year statements in the 10K, and quarterly filings are often a lighter cut than the annual filing. For the daily frequency signals (Volatility, Momentum, and the price five trading days before each earnings announcement used in Earnings Surprise and SUE), we use trading days only, skipping weekends and US market holidays. Volatility is reported as the standard deviation of daily returns, not annualized, to stay consistent with Part

1. For Momentum, Part 1 does not specify the market benchmark, and we used the S&P 500 (via the SPY ETF) as a standard US large cap proxy.

In Part 1 every signal had been standardized within each calendar month across 200 firms, so a value of 1 meant “one standard deviation above the average firm that month.” With only three firms in Part 2 a within-month z -score is not informative, so we report raw values in Table 5 and use within-trio ranks in Step B (see table note). The rest of the deviations from Part 1 definitions are documented inline above.

Step B: Combined Signal Analysis

Part 1 gives us a multivariate coefficient and a t statistic for each signal. We treat those coefficients as our beliefs about how characteristics map into expected return in the simulated world. For Part 2 we only have three firms, so we do not re estimate a cross sectional regression. Instead we ask, for each signal separately, which of the three firms looks best for a long position given (i) the sign of the Part 1 coefficient, (ii) how much we trust that signal, and (iii) the raw signal value from Table 5.

For reliability we give each signal a weight proportional to its absolute t statistic from the Part 1 multivariate step. Larger $|t|$ means more weight. The weights sum to one. They are reported in Table 6. Momentum, volatility, and insider selling receive the most weight because their $|t|$ statistics are largest in absolute value.

For each signal we convert the three raw values into rank scores within the trio. The best firm for that signal gets score 1, the middle firm 0.5, and the worst firm 0. If two firms tie, they share the average of the ranks they would have had, so a tie for first among two stocks gives each a score of 0.75 and the third stock scores 0.

The direction rule follows the multivariate sign. If the Part 1 coefficient is positive, a higher raw value is better for a long (we rank higher values toward the top). If the coefficient is negative, a lower raw value is better for a long (we rank lower values toward the top). That way a deep value name that is good on book to market is rewarded when book to market has a positive coefficient, and a firm with little insider selling is rewarded when insider selling has a negative coefficient.

The composite score for firm i is $\sum_j w_j s_{i,j}$ where w_j is the weight for signal j and $s_{i,j}$ is the rank score for firm i on signal j . We implement the arithmetic in `Code/07_stepB_composite_score.py`, which writes summary output to `DATA/stepB_composite_scores.csv`.

Table 6. Step B weights: proportional to $|t|$ from Part 1 multivariate regression.

Signal	$ t $	Weight
inssell	6.3052	0.1487
booktomarket	2.8890	0.0682
momentum	7.7881	0.1837
grossprofit	3.2374	0.0764
earnssurprise	5.5437	0.1308
sue	2.4416	0.0576
fscore	2.8494	0.0672
xfin	4.7232	0.1114
volatility	6.6122	0.1560
Sum	42.3898	1.0000

Table 7. Rank scores by signal before weighting (1 is best for a long given the Part 1 sign, 0.5 is middle, 0 is worst; ties averaged).

Signal	EL	ELF	COTY
inssell	0.50	0.00	1.00
booktomarket	0.00	0.50	1.00
momentum	1.00	0.50	0.00
grossprofit	0.50	1.00	0.00
earnssurprise	0.50	1.00	0.00
sue	1.00	0.50	0.00
fscore	0.75	0.75	0.00
xfin	0.50	1.00	0.00
volatility	1.00	0.00	0.50

Table 8. Composite scores and recommendations.

Company	Composite	Rank	Recommendation
Estée Lauder (EL)	0.6814	1	Buy
e.l.f. Beauty (ELF)	0.5237	2	Neutral
Coty (COTY)	0.2949	3	Sell

Investment recommendations

Estée Lauder is not the best name on every line item, but it collects enough mass on the signals that Part 1 weights most heavily. Momentum carries the largest single weight in our blend, and EL has the strongest twelve month market adjusted momentum among the three. That lines up with a large positive multivariate coefficient on momentum in Part 1. Volatility is the next heaviest weight among negative beta signals, and EL also has the lowest daily volatility in our sample. Low risk names that still carry positive trend are naturally attractive when the model associates lower volatility with higher return and rewards continuation.

EL also leads on SUE, another positively weighted signal, and earns middling credit on book to market, gross profitability, and earnings surprise rather than leading them. The F Score tie with ELF gives partial rather than full credit on that line, which is appropriate because a tie means we should not treat one of the two as clearly safer on Piotroski terms. On external financing EL sits between Coty and ELF. EL is not the standout on xfin in our snapshot, but it is not punished as heavily as Coty on the signals that pull the composite down for the third place name.

Netting this out, Estée Lauder looks like the most balanced way to express our Part 1 beliefs inside a three stock beauty sleeve as of late April 2026. It is the only company that simultaneously posts acceptable readings on the high weight positive themes (momentum, SUE) and on the high weight risk themes (volatility, and middling insider selling rather than the worst print in the group).

ELF scores very well on gross profitability, which matters in the model because gross profitability has a positive and statistically reliable multivariate slope. ELF also wins on earnings surprise and on external financing in our ranking exercise, which matches negative coefficients on xfin in Part 1 where lower issuance pressure is treated as better for a long in this construction. Those strengths pull ELF clearly above Coty.

At the same time ELF is weak on momentum relative to EL and is the worst of the three on insider selling in our filtered Form 4 window and the worst on volatility. Momentum and

volatility receive more weight than gross profitability in our scheme because their absolute t statistics are larger. That downweights ELF's profitability edge just enough to leave it in second place rather than first. The Neutral label therefore means we would not lean short the name under our quantitative rules, but we would also not make it the top pick ahead of EL unless we changed weights or added a discretionary override discussed below. If forced into a directional position rather than a true zero, ELF is a marginal long given its grossprofit and xfin strength, but at materially smaller size than EL.

Coty's score is dragged down by momentum, earnings surprise, and SUE, where it ranks last among the three. Those three signals all enter with positive Part 1 coefficients, so trailing price behavior and earnings trajectory matter a great deal. Coty's high book-to-market and the best insider-flow rank in our window are not enough to salvage the composite, because together those two signals receive less weight than momentum alone, and Coty still posts the weakest momentum print by a wide margin.

In plain terms, the model that we calibrated in Part 1 reads Coty's current signal stack as more distressed and more backward looking on price and earnings trends, even if the balance sheet looks cheap on a ratio basis. Our Sell recommendation is therefore an output of consistency with Part 1, not a claim that Coty can never rerate if operating results inflect.

Trailing equity performance through early 2026 has been most supportive for ELF's growth story, mixed-to-improving for EL as operating measures inflect, and weakest for Coty after a sharp fiscal reset and heavy selling pressure into calendar spring. That pattern is exactly what a model dominated by momentum and earnings-trajectory inputs would emphasize, so we treat the composite as a medium-term (order one year) factor-relative read, not a decades-long structural bet on book ratios alone. Long EL versus short COTY is a within-beauty relative expression; it does not hedge broader consumer discretionary or market beta.

This is a rank exercise among three related consumer names, not a full portfolio optimization. We do not size positions, hedge currency, or model transaction costs here. Sector concentration is total because the trio is already in the same industry. Anyone implementing a real trade would still want liquidity checks, borrow availability for a Sell, and a view on whether the April 2026 signal snapshot is stale after events.

To verify that the recommendation does not hinge on a single weighting choice, Table 9 recomputes the composite under five alternative schemes: equal weights across the nine signals, implied-Sharpe weights from Step C (which are proportional to $|t|/\sqrt{T}$ and therefore monotone in $|t|$), and three leave-one-out perturbations that drop in turn the heaviest positive signal (momentum), the heaviest negative signal (volatility), and the highest-judgment-call signal (insider selling, given the Coty division-president and ELF option-exercise calls described in Step A). The Buy / Neutral / Sell ordering is invariant in every case: EL is first on all six rows, COTY last on all six.

The sensitivity is informative on margins as well as on the ranking. Dropping momentum compresses the EL–COTY gap from 0.39 to 0.25, confirming that momentum is the single most important driver of the spread without being the only thing keeping COTY at the bottom. Dropping volatility nearly equalizes EL and ELF (0.6225 vs. 0.6205), which is the cleanest test of whether EL's lead is a low-volatility premium in disguise: it is partly that, but EL still edges ELF on positive-loading signals (momentum, SUE) so the order does not flip. Equal-weighting keeps the ranking but tightens the spread, as expected when high-conviction signals are downweighted relative to noisier ones. Implied-Sharpe-weighting produces identical scores

Table 9. Composite score sensitivity under alternative weighting schemes.

Scheme	EL	ELF	COTY	Order (high $\rightarrow$ low)
$ t $ -weighted (baseline)	0.6814	0.5237	0.2949	EL, ELF, COTY
Equal-weighted	0.6389	0.5833	0.2778	EL, ELF, COTY
Implied-Sharpe-weighted	0.6814	0.5237	0.2949	EL, ELF, COTY
Drop momentum	0.6097	0.5291	0.3613	EL, ELF, COTY
Drop volatility	0.6225	0.6205	0.2570	EL, ELF, COTY
Drop insider selling	0.7131	0.6152	0.1717	EL, ELF, COTY

to the $|t|$ -weighted baseline because both weight schemes are monotone transformations of $|t|$ when the number of months T is the same for every signal, which it is here.

A concrete illustration of what would have to happen for the ranking to break: with momentum carrying nearly a fifth of total weight in the baseline, if Coty improved from last to first on that rank while EL dropped to the middle, the EL–COTY composite gap would close by roughly half before touching any other line item. Pair that kind of shift with similar moves on earnings surprise and SUE—where Coty also ranks last today—and the stack could in principle reorder. The point is that the short is anchored in trend and revision inputs, not in a standalone balance-sheet judgment, but the leg most exposed to a regime change is the Coty short rather than the EL long.

Discretionary analysis

The factor stack omits operating narrative; the lines below lean on issuer disclosure and desk news through May 2026, with sources at first mention.

If we ask where discretion pushes hardest against the quant output, it is sizing the Coty short, not tearing up the EL long. EL’s fiscal Q3 2026 print (modest organic growth, materially higher adjusted EPS, and a raised FY26 outlook¹) supports the long, while travel-retail timing in Asia and Middle East disruption flagged for Q4 sit outside our signals. Coty already repriced on a fiscal Q2 2026 loss, like-for-like revenue pressure, withdrawn full-year guidance, and the “Coty. Curated.” refocus under interim leadership.² Factor momentum remains awful there, but narrative and price make that the leg a PM would scale down first.

The cleanest blind spot in our stack is geopolitical. ELF sources the overwhelming majority of its products from China per its FY25 10-K, leaving FY26 gross margin highly exposed to U.S. tariff escalation; this is not captured by any factor we use, sits squarely against ELF’s gross-profitability strength in our composite, and is more asymmetric than the May Morgan Stanley equal-weight we already see in price³. EL’s analogous exposure is travel-retail concentration in Asia, the structural backdrop to its \$1.5–\$1.7 billion BRP restructuring rather than a one-quarter item.

¹The Estée Lauder Companies, *Fiscal 2026 Third Quarter Results*, company press release, 1 May 2026, <https://www.elcompanies.com/en/news-and-media/newsroom/press-releases/2026/05-01-2026-110042145>.

²Coty Inc., *Second Quarter Fiscal Year 2026 Results*, 5 February 2026, <https://www.coty.com/news/coty-announces-second-quarter-fiscal-year-2026-results>; Marie Lampert and Siddarth S, “Coty shifts focus to core brands under new CEO, withdraws full-year outlook,” *Reuters*, 5 February 2026, <https://www.reuters.com/business/retail-consumer/coty-shifts-focus-core-brands-under-new-ceo-withdraws-full-year-outlook-2026-02-05/>.

³“Morgan Stanley downgrades e.l.f. Beauty stock rating on market share losses,” *Investing.com*, 1 May 2026, <https://www.investing.com/news/analyst-ratings/morgan-stanley-downgrades-elf-beauty-stock-rating-on-market-share-losses-93CH-4651891>.

A second blind spot is methodological. Our Part 1 book-to-market slope is a partial effect identified off within-month dispersion across 200 firms; the COTY–EL B/M spread is $\sim 13\times$, an out-of-sample extrapolation. A quant fund should winsorize extreme single-name characteristics before applying pooled coefficients; a discretionary fund would re-underwrite COTY’s balance sheet directly. The composite still ranks COTY last on momentum and volatility, but this is where quant and discretionary processes diverge in implementation even when they agree on direction.